



Using Pull Requests to Make Collaboration Visible in CDIO Project-Based Courses

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INDUSTRIAL ENGINEERING

Project-based courses

Assessment often prioritizes final artifacts, limiting visibility into collaboration, reasoning, and responsiveness to feedback

Professional practice

Modern technical work relies on version control, peer review, and documented iteration—skills students are expected to use immediately in the workforce

Assessment misalignment

What we value in learning is not always what we can currently observe

Core question

How can collaboration, accountability, and responsiveness to feedback be made **visible and assessable** in team-based projects?

- ▶ final deliverables hide process and contribution
- ▶ peer interaction is difficult to audit retrospectively
- ▶ feedback often resets instead of accumulating



Not just another tool

Git/GitHub is an industry-standard collaboration infrastructure

- ▶ used across software, data science, and engineering teams
- ▶ supports version control, review, and reproducible handover
- ▶ directly transferable – students can credibly list this on their CV



Motivation: students practice skills they will immediately use

Repository

A shared workspace containing code, data, documentation, and history

Pull Request (PR)

A structured proposal to change shared work, including:

- ▶ rationale and scope
- ▶ peer review and discussion
- ▶ explicit approval and merge decision

 **Key idea: PRs bind technical change to collaborative process**

Starting point

- ▶ CDIO-aligned
- ▶ assessment as development
- ▶ observable collaborative

Design claim

Pull requests are not treated as a tooling detail, but as **assessment infrastructure**:

 feedback +  iteration $\Rightarrow$  competence

Repository

A shared workspace containing code, data, documentation, and history

Pull Request (PR)

A structured proposal to change shared work through:

- ▶ rationale and scope
- ▶ peer review and discussion
- ▶ approval and merge decision

Why it matters for assessment

- ▶ reasoning becomes visible
- ▶ contribution becomes attributable
- ▶ critique and negotiation are captured
- ▶ responses to feedback are auditable

Reframing

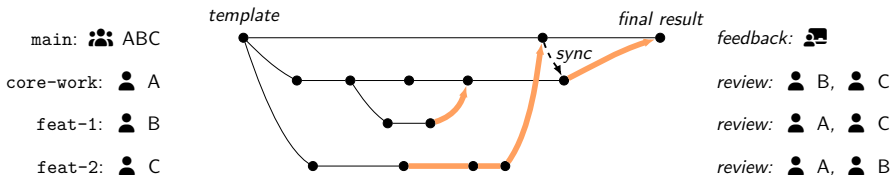
PRs expose the learning process

Required practices

- ▶ focused PRs with clear intent
- ▶ at least two substantive reviews
- ▶ explicit responses to all comments
- ▶ merge only after review

Observable evidence

- ▶ authorship and co-authorship
- ▶ depth of critique and follow-up
 -  actionable comments (no LGTM or )
- ▶ handover quality
- ▶ documentation updates



Information Engineering (IE)

- ▶ 20–25 students
- ▶ 6–7 small teams
- ▶ early undergraduate
- ▶ new repository each cycle

Main purpose

Tooling and collaboration practices are the learning focus

Business Intelligence (BI)

- ▶ 10–15 students
- ▶ 2 large teams
- ▶ assumes GitHub fluency
- ▶ single persistent repository

Main purpose

Tooling supports more advanced project work

Cumulative assessment

- ▶ unresolved PR comments remain active
- ▶ later work is checked against prior feedback
- ▶ capstone evaluation includes earlier issues

Effect

- ▶ continuity
- ▶ accountability
- ▶ coherence
- ▶ less fragmentation

	Outstanding	Good	Developmental
GitHub usage	Branches are well organized with logical names tied to features. Commits are coherent and clearly describe changes. Documentation is complete and provides clear instructions.	Branches are used but naming or scope is sometimes inconsistent. Commits are mostly coherent with occasional issues. Documentation is present but incomplete.	Branches are absent or unclear. Commits are inconsistent with little structure. Documentation is unclear or missing.
Teamwork	All team members actively contribute. Contributions and responsibilities are distributed across the team. Decisions are shared through review and discussion.	Most members participate, but contributions are uneven. One or two members lead development, while others contribute through reviews or comments.	Work is largely carried out by one individual. Participation is minimal or uneven across the team.

Calibration

- ▶ levels: developmental → outstanding
- ▶ same dimensions, increasing expectations
- ▶ signals expectations for Capstone

Observed trend

- ▶ GitHub usage improved over repeated cycles
- ▶ teamwork & participation improved more strongly

Balancing team and individual assessment

Team-level

- ▶ analytical quality
- ▶ coherence
- ▶ shared responsibility

Individual-level

- ▶ visible contributions
- ▶ review quality
- ▶ responsiveness to feedback

Design goal

Accountability without fragmenting teamwork

Key shifts in practice



Quality

iteration
vs. reviewable work



Collaboration

implicit teamwork
vs. explicit review



Workflow

informal work
vs. structured process



Takeaway

Making collaboration visible turns implicit practices into something we can teach and assess



Boundary conditions

- ▶ design-oriented, not experimental
- ▶ depends on student readiness
- ▶ larger cohorts require more coordination



Next iterations

- ▶ extend to more courses
- ▶ structured team-building to accelerate trust
- ▶ refined review calibration

Key takeaway

PR-based assessment makes collaboration, accountability, and responsiveness to feedback visible and assessable

- ▶ assessment moves from final products to visible process
- ▶ feedback carries forward instead of resetting
- ▶ students practice professional collaboration in routine course work

Contribution

This is an assessment architecture, not just a tool choice

Questions?



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